

Optimal Tests in Higher Dimensions

In Section 13.2, uniformly most powerful unbiased tests are considered for multiparameter exponential families. The discussion involves marginal and conditional distributions described in Section 13.1. The t -test and Fisher's exact test are considered as examples in Section 13.3.

13.1 Marginal and Conditional Distributions

The main result of this section uses the following technical lemma about conditional and marginal distributions when a joint density factors, but the dominating measure is not a product measure.

Lemma 13.1. *Let Y be a random vector in $\mathbb{R}^n$ and T a random vector in $\mathbb{R}^m$, and let P_0 and P_1 be two possible joint distributions for Y and T . Introduce marginal distributions*

$$Q_0(B) = P_0(T \in B) \text{ and } Q_1(B) = P_1(T \in B),$$

and conditional distributions

$$R_{0t}(B) = P_0(Y \in B | T = t) \text{ and } R_{1t}(B) = P_1(Y \in B | T = t).$$

Assume $P_1 \ll P_0$ and that the density for P_1 has form

$$\frac{dP_1}{dP_0}(t, y) = a(y)b(t)$$

with $a(y) > 0$ for all $y \in \mathbb{R}^n$. Then $Q_1 \ll Q_0$ and $R_{1t} \ll R_{0t}$, for a.e. t (Q_1) with densities given by

$$\frac{dQ_1}{dQ_0}(t) = b(t)E_0[a(Y)|T = t] = b(t) \int a \, dR_{0t}$$

and

$$\frac{dR_{1t}}{dR_{0t}}(y) = \frac{a(y)}{\int a \, dR_{0t}}.$$

Proof. The formula for the marginal density is established by first using Lemma 12.18 to write $P_1(T \in B)$ as an expectation under P_0 involving the likelihood ratio $a(Y)b(T)$, followed by a smoothing argument to write this expectation as an integral against Q_0 . Thus

$$\begin{aligned} P_1(T \in B) &= E_0[I\{T \in B\}a(Y)b(T)] \\ &= E_0 E_0[I\{T \in B\}a(Y)b(T) \mid T] \\ &= E_0[I\{T \in B\}b(T)E_0[a(Y) \mid T]] \\ &= \int_B b(t)E_0[a(Y) \mid T = t]dQ_0(t). \end{aligned}$$

Next, if the stated density for R_{1t} is correct,

$$R_{1t}(C) = \int_C \frac{a(y)}{\int a dR_{0t}} dR_{0t}(y),$$

and so, according to Definition 6.2 of conditional distributions, we must show that

$$P_1(T \in B, Y \in C) = \int_B \int_C \frac{a(y)}{\int a dR_{0t}} dR_{0t}(y) dQ_1(t).$$

Using the formula for the density of Q_1 with respect to Q_0 , the right-hand side of this equation equals

$$\begin{aligned} &\int_B \int I\{y \in C\}a(y) dR_{0t}(y)b(t) dQ_0(t) \\ &= E_0[I\{T \in B\}b(T)E_0[I\{Y \in C\}a(Y) \mid T]] \\ &= E_0[a(Y)b(T)I\{T \in B, Y \in C\}] \\ &= P_1(T \in B, Y \in C), \end{aligned}$$

where the last equality follows from Lemma 12.18. $\square$

Suppose the distribution for data X comes from an $(r + s)$ -parameter canonical exponential family with densities

$$p_{\theta, \eta}(x) = h(x) \exp[\theta \cdot U(x) + \eta \cdot T(x) - A(\theta, \eta)], \quad (13.1)$$

where θ is r -dimensional and η is s -dimensional. The following theorem gives the form of marginal and conditional distributions for the sufficient statistics U and T .

Theorem 13.2. *If X has density $p_{\theta, \eta}$ in (13.1), then there exist measures λ_θ and ν_t such that:*

1. *With θ fixed, the marginal distributions of T will form an s -parameter exponential family with densities*

$$\exp[\eta \cdot t - A(\theta, \eta)]$$

with respect to λ_θ .

2. The conditional distributions of U given $T = t$ form an exponential family with densities

$$\exp[\theta \cdot u - A_t(\theta)]$$

with respect to ν_t . These densities are independent of η .

Proof. Fix some point $(\theta_0, \eta_0) \in \Omega$, and let ν be the joint distribution of T and U under P_{θ_0, η_0} . Arguing as in the proof of Theorem 12.19, under $P_{\theta, \eta}$, T and U have joint density

$$\exp[(\theta - \theta_0) \cdot u + (\eta - \eta_0) \cdot t + A(\theta_0, \eta_0) - A(\theta, \eta)]$$

with respect to ν . If R_0 denotes the conditional distribution of U given T under P_{θ_0, η_0} , then by Lemma 13.1 the marginal density of T under $P_{\theta, \eta}$ is

$$e^{\eta \cdot t - A(\theta, \eta)} \int \exp((\theta - \theta_0) \cdot u - \eta_0 \cdot t + A(\theta_0, \eta_0)) dR_{0t}(u)$$

with respect to Q_0 , the marginal density of T under P_{θ_0, η_0} . This is of the correct form provided we choose λ_θ so that

$$\frac{d\lambda_\theta}{dQ_0}(t) = \int \exp((\theta - \theta_0) \cdot u - \eta_0 \cdot t + A(\theta_0, \eta_0)) dR_{0t}(u).$$

By the second formula in Lemma 13.1, under $P_{\theta, \eta}$ the conditional density of U given $T = t$ with respect to R_{0t} is

$$\frac{e^{(\theta - \theta_0) \cdot u}}{\int e^{(\theta - \theta_0) \cdot v} dR_{0t}(v)}.$$

This density has the desired form. □

13.2 UMP Unbiased Tests in Higher Dimensions

If the power function β_φ for an unbiased test φ is continuous, then $\beta_\varphi(\theta) \leq \alpha$ for θ in $\overline{\Omega}_0$, the closure of Ω_0 , and $\beta_\varphi(\theta) \geq \alpha$ for $\theta \in \overline{\Omega}_1$. If we take $\omega = \overline{\Omega}_0 \cap \overline{\Omega}_1$, the common boundary of Ω_0 and Ω_1 , then

$$\beta_\varphi(\theta) = \alpha, \quad \forall \theta \in \omega.$$

Tests satisfying this equation are called α -similar. Here α need not denote the level of the tests, because β_φ may exceed α at points θ in $\Omega_0 - \omega$.

Lemma 13.3. *Suppose φ^* is α -similar and has level α , and that power functions β_φ for all test functions φ are continuous. If φ^* is uniformly most powerful among all α -similar tests, then it is unbiased and uniformly most powerful among all unbiased tests.*

Proof. The degenerate test that equals α regardless of the observed data is α -similar. Since φ^* has better power, $\beta_{\varphi^*}(\theta) \geq \alpha$, $\theta \in \Omega_1$. Because φ^* has level α , $\beta_{\varphi^*}(\theta) \leq \alpha$, $\theta \in \Omega_0$. Thus φ^* is unbiased. If a competing test φ is unbiased, then since its power function is continuous it is α -similar. Then $\beta_{\varphi} \leq \beta_{\varphi^*}$ on Ω_1 because φ^* is uniformly most powerful among all α -similar tests. $\square$

The tests we develop use conditioning to reduce to the univariate case. Part of why this works is that the tests have the structure in the following definition.

Definition 13.4. Suppose T is sufficient for the subfamily $\{P_{\theta} : \theta \in \omega\}$. An α -similar test φ has Neyman structure if

$$E_{\theta}[\varphi|T = t] = \alpha, \quad \text{for a.e. } t, \quad \forall \theta \in \omega.$$

Theorem 13.5. If T is complete and sufficient for $\{P_{\theta} : \theta \in \omega\}$, then every similar test has Neyman structure.

Proof. For $\theta \in \omega$, let $h(T) = E_{\theta}(\varphi|T)$. (Because T is sufficient, h is independent of $\theta \in \omega$.) By smoothing,

$$E_{\theta}\varphi = E_{\theta}h(T) = \alpha, \quad \forall \theta \in \omega.$$

By completeness, $h(T) = E_{\theta}(\varphi|T) = \alpha$, a.e., for all $\theta \in \omega$. $\square$

Suppose data X has density

$$p_{\theta, \eta}(x) = h(x) \exp\{\theta U(x) + \eta \cdot T(x) - A(\theta, \eta)\}. \quad (13.2)$$

Here θ is univariate, but η can be s -dimensional. The tests of interest are derived by conditioning on T . By Theorem 13.2, the conditional distributions of U given $T = t$ form a one-parameter exponential family with canonical parameter θ . Theorem 12.9 gives a uniformly most powerful conditional test of $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$, given by

$$\varphi_1 = \begin{cases} 1, & U > c(T); \\ \gamma(T), & U = c(T); \\ 0, & U < c(T), \end{cases}$$

with $c(\cdot)$ and $\gamma(\cdot)$ adjusted so that

$$P_{\theta_0, \eta}(U > c(t) \mid T = t) + \gamma(t)P_{\theta_0, \eta}(U = c(t) \mid T = t) = \alpha.$$

Similarly, Theorem 12.26 gives a uniformly most powerful unbiased conditional test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$, given by

$$\varphi_2 = \begin{cases} 1, & U < c_-(T); \\ 1, & U > c_+(T); \\ \gamma_-(T), & U = c_-(T); \\ \gamma_+(T), & U = c_+(T); \\ 0, & U \in (c_-(T), c_+(T)), \end{cases}$$

with $c_{\pm}(\cdot)$ and $\gamma_{\pm}(\cdot)$ adjusted so that

$$E_{\theta_0, \eta}[\varphi_2 | T = t] = \alpha$$

and

$$E_{\theta_0, \eta}[\varphi_2 U | T = t] = \alpha E_{\theta_0, \eta}[U | T = t].$$

Theorem 13.6. *If the exponential family (13.2) is of full rank and Ω is open, then φ_1 is a uniformly most powerful unbiased test of $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$, and φ_2 is a uniformly most powerful unbiased test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.*

Proof. Let us begin proving the assertion about φ_1 . First note that the conditions on the exponential family ensure that the densities with $\theta = \theta_0$ form a full rank exponential family with T as a complete sufficient statistic. Also, from the construction, $E_{\theta_0, \eta}[\varphi_1 | T] = \alpha$, so by smoothing $E_{\theta_0, \eta}\varphi_1 = \alpha$ and φ_1 is α -similar. Suppose φ is a competing α -similar test. Then φ has Neyman structure by Theorem 13.5 and $E_{\theta_0, \eta}[\varphi | T = t] = \alpha$. Because φ_1 is the most powerful conditional test of $\theta = \theta_0$ versus $\theta > \theta_0$, if $\theta > \theta_0$, then

$$E_{\theta, \eta}(\varphi_1 | T = t) \geq E_{\theta, \eta}(\varphi | T = t),$$

and by smoothing,¹

$$E_{\theta, \eta}\varphi_1 = E_{\theta, \eta}E_{\theta, \eta}(\varphi_1 | T) \geq E_{\theta, \eta}E_{\theta, \eta}(\varphi | T) = E_{\theta, \eta}\varphi.$$

This shows that φ_1 is uniformly most powerful α -similar. By Theorem 12.9, the conditional power function for φ_1 is increasing in θ , and so if $\theta < \theta_0$,

$$E_{\theta, \eta}\varphi_1 = E_{\theta, \eta}E_{\theta, \eta}(\varphi_1 | T) \leq E_{\theta_0, \eta}E_{\theta, \eta}(\varphi_1 | T) = \alpha.$$

Thus φ_1 has level α . By Theorem 2.4, power functions for all test functions are continuous, so by Lemma 13.3 φ_1 is uniformly most powerful unbiased.

The argument for the assertion about φ_2 is a bit more involved. Let

$$m(\theta, \eta) = E_{\theta, \eta}U = \frac{\partial A(\theta, \eta)}{\partial \eta},$$

by (2.4). By dominated convergence (as in Theorem 12.20),

¹ There is a presumption here that (θ_0, η) lies in Ω regardless the choice of $(\theta, \eta) \in \Omega$. Unfortunately, this does not have to be the case. This issue can be resolved by reparameterization. If $(\theta_0, \eta_0) \in \Omega$ and we are concerned with power at $(\theta_1, \eta_1) \in \Omega$, define new parameters $\tilde{\theta} = \theta$ and $\tilde{\eta} = \eta + (\eta_0 - \eta_1)(\theta - \theta_0)/(\theta_1 - \theta_0)$. Then the original parameters of interest, (θ_0, η_0) and (θ_1, η_1) become $(\tilde{\theta}_0, \tilde{\eta}_0) = (\theta_0, \eta_0)$ and $(\tilde{\theta}_1, \tilde{\eta}_1) = (\theta_1, \eta_0)$. The canonical statistics for the reparameterized family are $\tilde{T} = T$ and $\tilde{U} = U + (\eta_0 - \eta_1) \cdot T/(\theta_1 - \theta_0)$. Since we condition on T , it is easy to see that the test φ_1 will be the same, regardless of the parameterization.

$$\begin{aligned}
\frac{\partial \beta_\varphi(\theta, \eta)}{\partial \theta} &= \frac{\partial}{\partial \theta} \int \varphi(x) p_{\theta, \eta}(x) d\mu(x) \\
&= \int \varphi(x) \frac{\partial}{\partial \theta} p_{\theta, \eta}(x) d\mu(x) \\
&= \int \varphi(x) (U(x) - m(\theta, \eta)) p_{\theta, \eta}(x) d\mu(x) \\
&= E_{\theta, \eta} \varphi (U - m(\theta, \eta)).
\end{aligned}$$

Suppose φ is unbiased. Then this derivative must be zero when $\theta = \theta_0$, and thus

$$E_{\theta_0, \eta} \varphi U - \alpha m(\theta_0, \eta) = E_{\theta_0, \eta} [\varphi U - \alpha U] = 0.$$

Conditioning on T ,

$$0 = E_{\theta_0, \eta} E_{\theta_0, \eta} [\varphi U - \alpha U \mid T],$$

and since T is complete for the family of distributions with $\theta = \theta_0$, this implies that

$$E_{\theta_0, \eta} [\varphi U - \alpha U \mid T] = 0.$$

But φ is α -similar and has Neyman structure, implying $E_{\theta_0, \eta} [\varphi | T] = \alpha$, and so

$$E_{\theta_0, \eta} [\varphi U | T] = \alpha E_{\theta_0, \eta} [U | T].$$

By Theorem 12.20, this constraint ensures that the conditional power $E_{\theta, \eta} [\varphi | T]$ has zero slope at $\theta = \theta_0$. By Theorem 12.22, φ_2 is the uniformly most powerful conditional test satisfying this condition, and so

$$E_{\theta, \eta} [\varphi_2 | T] \geq E_{\theta, \eta} [\varphi | T].$$

Taking expectations, by smoothing

$$E_{\theta, \eta} \varphi_2 \geq E_{\theta, \eta} \varphi.$$

Thus φ is uniformly most powerful unbiased. (Again, reparameterization can be used to treat cases where $(\theta, \eta) \in \Omega$ but $(\theta_0, \eta) \notin \Omega$.) $\square$

13.3 Examples

Example 13.7 (The t -test). The theory developed in the previous section can be used to test the mean of a normal distribution. Suppose $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$, and consider testing $H_0 : \mu \leq 0$ versus $H_1 : \mu > 0$. The joint density from Example 2.3 is

$$\frac{1}{(2\pi)^{n/2}} \exp \left[\frac{\mu}{\sigma^2} U(x) - \frac{1}{2\sigma^2} T(x) - \frac{n\mu^2}{2\sigma^2} - n \log \sigma \right],$$

with $U(x) = x_1 + \cdots + x_n$ and $T(x) = x_1^2 + \cdots + x_n^2$. This has form (13.2) with $\theta = \mu/\sigma^2$ and $\eta = -1/(2\sigma^2)$. Note that the hypotheses expressed using the canonical parameters are $H_0 : \theta \leq 0$ versus $H_1 : \theta > 0$. By Theorem 13.6, the uniformly most powerful unbiased test has form

$$\varphi = \begin{cases} 1, & U > c(T); \\ 0, & \text{otherwise,} \end{cases}$$

with $c(\cdot)$ chosen so that

$$P_{\mu=0}[U > c(t) \mid T = t] = \alpha.$$

To proceed we need the conditional distribution of U given $T = t$ when $\mu = 0$. Note that the family of distributions with $\mu = 0$ is an exponential family with complete sufficient statistic T . Also, if we define $Z = X/\sigma$, so that $Z_1, \dots, Z_n$ are i.i.d. standard normal, or $Z \sim N(0, I)$, then $W = X/\|X\| = Z/\|Z\|$ is ancillary. By Basu's theorem (Theorem 3.21), T and W are independent. Because $\|X\| = \sqrt{T}$, $X = W\sqrt{T}$, and using independence between T and W ,

$$E[h(X)|T = t] = E[h(W\sqrt{t})|T = t] = Eh(W\sqrt{t}).$$

This shows that

$$X|T = t \sim W\sqrt{t}. \quad (13.3)$$

The vector W is said to be uniformly distributed on the unit sphere. Note that if O is an arbitrary orthogonal matrix ($OO' = I$), then $OZ \sim N(0, OO') = N(0, I)$. Also $\|OZ\|^2 = (OZ)'(OZ) = Z'O'OZ = Z'Z = \|Z\|^2$. Thus Z and OZ have the same length and distribution. Then

$$OW = \frac{OZ}{\|Z\|} = \frac{OZ}{\|OZ\|} \sim \frac{Z}{\|Z\|} = W.$$

So W and OW have the same distribution, which shows that the uniform distribution on the unit sphere is invariant under orthogonal transformations. In fact, this is the only probability distribution on the unit sphere that is invariant under orthogonal transformations. Using (13.3), since $U = \mathbf{1}'X$ where $\mathbf{1}$ denotes a column of 1s,

$$P_{\mu=0}[U > c(t) \mid T = t] = P(\mathbf{1}'W > c(t)/\sqrt{t}).$$

This equals α if we take $c(t)/\sqrt{t} = q$, the upper α th quantile for $\mathbf{1}'W$. Thus the uniformly most powerful unbiased test rejects H_0 if

$$\frac{U}{\sqrt{T}} \geq q.$$

Although it may not be apparent, this is equivalent to the usual test based on the t -statistic

$$t = \frac{\bar{X}}{S/\sqrt{n}}.$$

To see this, note that

$$\bar{X} = U/n$$

and

$$\begin{aligned} S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n X_i^2 - \frac{n}{n-1} \bar{X}^2 = \frac{T}{n-1} - \frac{U^2}{n(n-1)}, \end{aligned}$$

and so

$$t = \frac{U/\sqrt{n}}{\sqrt{(T - U^2/n)/(n-1)}} = \frac{\sqrt{n-1} \operatorname{Sign}(U)}{\sqrt{nT/U^2 - 1}} = g(U/\sqrt{T}).$$

The function $g(\cdot)$ here is strictly increasing, and so $U/\sqrt{T} > q$ if and only if $t > g(q)$. When $\mu = 0$, t has the t -distribution on $n-1$ degrees of freedom, and so level α is achieved taking $g(q) = t_{\alpha, n-1}$, the upper α th quantile of this distribution. So our test then rejects H_0 when

$$t > t_{\alpha, n-1}.$$

Details for the two-sided case, testing $H_0 : \mu = 0$ versus $H_1 : \mu \neq 0$, are similar. The uniformly most powerful level α test rejects H_0 when $|t| > t_{\alpha/2, n-1}$.

Example 13.8 (Fisher's Exact Test). A second example of unbiased testing concerns dependence in a two-way contingency table. Consider two questions on a survey, A and B , and suppose each of these questions has two answers. Responses to these questions might be coded with variables $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ taking $X_k = 1$ if respondent k gives the first answer to question A , $X_k = 2$ if respondent k gives the second answer to question A ; and $Y_k = 1$ if respondent k gives the first answer to question B , $Y_k = 2$ if respondent k gives the second answer to question B . If the pairs (X_k, Y_k) , $k = 1, \dots, n$, are i.i.d., and if

$$p_{ij} = P(X_k = i, Y_k = j), \quad i = 1, 2, \quad j = 1, 2,$$

then the joint density is

$$P(X_1 = x_1, \dots, X_n = x_n, Y_1 = y_1, \dots, Y_n = y_n) = \prod_{i=1}^2 \prod_{j=1}^2 p_{ij}^{n_{ij}}, \quad (13.4)$$

where $n_{ij} = \#\{k : x_k = i, y_k = j\}$. So if we take

$$N_{ij} = \#\{k : X_k = i, Y_k = j\}, \quad i = 1, 2, \quad j = 1, 2,$$

then $N = (N_{11}, N_{12}, N_{21}, N_{22})$ is a sufficient statistic. Based on these data, we may want to test whether there is positive dependence between the two questions. But first we need to resolve what we mean by “positive dependence.” There seem to be various possibilities.

Let (X, Y) be a generic variable distributed as (X_k, Y_k) . Perhaps we should define positive dependence between the questions to mean that the correlation between X and Y is positive. Because

$$E(X - 1) = P(X = 2) = p_{21} + p_{22} \stackrel{\text{def}}{=} p_{2+}$$

and

$$E(Y - 1) = P(Y = 2) = p_{12} + p_{22} \stackrel{\text{def}}{=} p_{+2},$$

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(X - 1, Y - 1) = E(X - 1)(Y - 1) - p_{2+}p_{+2} \\ &= P(X = 2, Y = 2) - p_{2+}p_{+2} = p_{22} - p_{2+}p_{+2} = p_{22}p_{11} - p_{12}p_{21}. \end{aligned}$$

So the covariance between X and Y is positive if and only if

$$p_{22}p_{11} > p_{12}p_{21}.$$

Another notion of positive dependence might be that the chance X equals the larger value 2 increases if we learn that Y equals its larger value, that is, if

$$P(X = 2|Y = 2) > P(X = 2).$$

Equivalently,

$$\frac{p_{22}}{p_{12} + p_{22}} > p_{21} + p_{22}.$$

Cross-multiplication and a bit of algebra show that this happens if and only if $p_{22}p_{11} > p_{12}p_{21}$, so this notion of positive dependence is the same as the notion based on correlation.

The distribution of N is multinomial,

$$P(N_{11} = n_{11}, \dots, N_{22} = n_{22}) = \binom{n}{n_{11}, \dots, n_{22}} p_{11}^{n_{11}} p_{12}^{n_{12}} p_{21}^{n_{21}} p_{22}^{n_{22}}.$$

It is convenient to introduce new variables $U = N_{11}$, $T_1 = N_{11} + N_{12}$, and $T_2 = N_{11} + N_{21}$. If N is given in a two-way table as in Example 5.5, then T_1 and T_2 determine the marginal sums. Given U and T , we can solve for N , specifically,

$$N_{11} = U, \quad N_{12} = T_1 - U, \quad N_{21} = T_2 - U,$$

and

$$N_{22} = n + U - T_1 - T_2.$$

Thus there is a one-to-one relation between N and variables T and U , and

$$\begin{aligned}
 P(T = t, U = u) &= P(N_{11} = u, N_{12} = t_1 - u, N_{21} = t_2 - u, N_{22} = n + u - t_1 - t_2) \\
 &= h(u, t) p_{11}^u p_{12}^{t_1 - u} p_{21}^{t_2 - u} p_{22}^{n + u - t_1 - t_2} \\
 &= h(u, t) \left(\frac{p_{11} p_{22}}{p_{12} p_{21}} \right)^u \left(\frac{p_{12}}{p_{22}} \right)^{t_1} \left(\frac{p_{21}}{p_{22}} \right)^{t_2} p_{22}^n \\
 &= h(u, t) \exp\{\theta u + \eta \cdot t - A(\theta, \eta)\}, \tag{13.5}
 \end{aligned}$$

where

$$\begin{aligned}
 \theta &= \log \left(\frac{p_{11} p_{22}}{p_{12} p_{21}} \right), \quad \eta = \begin{pmatrix} \log(p_{12}/p_{22}) \\ \log(p_{21}/p_{22}) \end{pmatrix}, \\
 h(u, t) &= \binom{n}{u, t_1 - u, t_2 - u, n + u - t_1 - t_2},
 \end{aligned}$$

and

$$A(\theta, \eta) = -n \log p_{22}.$$

Using Theorem 13.6, a uniformly most powerful unbiased test is given by

$$\varphi = \begin{cases} 1, & U > c(T); \\ \gamma(T), & U = c(T); \\ 0, & U < c(T), \end{cases}$$

with $c(\cdot)$ and $\gamma(\cdot)$ adjusted so that

$$\alpha = P_{\theta=0}(U > c(t)|T = t) + \gamma(t)P_{\theta=0}(U = c(t)|T = t).$$

To describe the test in a more explicit fashion, we need the conditional distribution of U given $T = t$ when $\theta = 0$. This distribution does not depend on η . It is convenient to assume that $p_{11} = p_{12} = p_{21} = p_{22} = 1/4$ and to denote probability in this case by P_0 . Then $P_0(X_k = 1) = P_0(X_k = 2) = P_0(Y_k = 1) = P_0(Y_k = 2) = 1/2$, so the joint density in (13.4) equals the product of the marginal densities and under P_0 the variables X_k , $k = 1, \dots, n$, and Y_k , $k = 1, \dots, n$, are all independent. Since T_1 depends on X_k , $k = 1, \dots, n$, and T_2 depends on Y_k , $k = 1, \dots, n$, T_1 and T_2 are independent under P_0 , each from a binomial distribution with n trials and success probability $1/2$. Thus

$$P_0(T = t) = P_0(T_1 = t_1)P_0(T_2 = t_2) = \binom{n}{t_1} \binom{n}{t_2} \frac{1}{4^n}.$$

Using (13.5),

$$P_0(U = u, T = t) = \binom{n}{u, t_1 - u, t_2 - u, n + u - t_1 - t_2} \frac{1}{4^n}.$$

Dividing these expressions, after a bit of algebra,

$$P_0(U = u|T = t) = \frac{P_0(U = u, T = t)}{P_0(T = t)} = \frac{\binom{t_1}{u} \binom{n-t_1}{t_2-u}}{\binom{n}{t_2}}.$$

This is the hypergeometric distribution, which arises in sampling theory. Consider drawing t_2 times without replacement from an urn containing t_1 red balls and $n - t_1$ white balls, and let U denote the number of red balls in the sample. Then there are $\binom{n}{t_2}$ samples, and the number of samples for which $U = u$ is $\binom{t_1}{u} \binom{n-t_1}{t_2-u}$. If the chances for all possible samples are the same, then the chance $U = u$ is given by the formula above.

The two-sided case can be handled in a similar fashion. Direct calculation shows that $\theta = 0$ if and only if X and Y are independent, and so testing $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$ amounts to testing whether answers for the two questions are independent. Again the best test conditions on the margins T , and probability calculations are based on the hypergeometric distribution. Calculations to set the constants $c_{\pm}$ and $\gamma_{\pm}$ are messy and need to be done numerically. These tests for two-way contingency tables were introduced by Fisher and are called *Fisher's exact tests*.

13.4 Problems²

*1. Consider a two-parameter exponential family with Lebesgue density

$$p_{\theta, \phi}(x, y) = (x + y)e^{\theta x + \phi y - A(\theta, \phi)}, \quad x \in (0, 1), \quad y \in (0, 1).$$

- Find $A(\theta, \phi)$.
 - Find the marginal density of X . Check that the form of this distribution agrees with Theorem 13.2.
 - Find the conditional density of X given $Y = y$. Again, check that this agrees with Theorem 13.2.
 - Determine the uniformly most powerful unbiased test of $H_0 : \theta \leq 0$ versus $H_1 : \theta > 0$.
 - Determine the uniformly most powerful unbiased test of $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$.
2. Suppose X and Y are absolutely continuous with joint density

$$p_{\theta, \eta}(x, y) = \begin{cases} \eta(\theta + \eta)e^{\theta x + \eta y}, & 0 < x < y; \\ 0, & \text{otherwise,} \end{cases}$$

where $\eta < 0$ and $\eta + \theta < 0$.

² Solutions to the starred problems are given at the back of the book.

- a) Determine the marginal density of Y . Show that for fixed θ these densities form an exponential family with parameter η .
- b) Determine the conditional density of X given $Y = y$. Show that for fixed y , these densities form an exponential family with parameter θ .
- *3. Let X and Y be independent variables, both with gamma distributions. The parameters for the distribution of X are α_x and λ_x ; the parameters for Y are α_y and λ_y ; and both shape parameters, α_x and α_y , are known constants.
 - a) Determine the uniformly most powerful unbiased test of $H_0 : \lambda_x \leq \lambda_y$ versus $H_1 : \lambda_x > \lambda_y$. Hint: You should be able to relate the critical value for the conditional test to a quantile for the beta distribution.
 - b) If $X_1, \dots, X_n$ is a random sample from $N(0, \sigma_x^2)$ and $Y_1, \dots, Y_m$ is a random sample from $N(0, \sigma_y^2)$, then one common test of $H_0 : \sigma_x^2 \leq \sigma_y^2$ versus $H_1 : \sigma_x^2 > \sigma_y^2$ rejects H_0 if and only if $F = s_x^2/s_y^2$ exceeds the upper α th quantile of the F -distribution on n and m degrees of freedom, where $s_x^2 = \sum_{i=1}^n X_i^2/n$ and $s_y^2 = \sum_{j=1}^m Y_j^2/m$. Show that this test is the same as the test in part (a). Give a formula relating quantiles for the F -distribution to quantiles for the beta distribution.
4. Consider a regression model in which data $Y_1, \dots, Y_n$ are independent with $Y_i \sim N(\alpha + \beta x_i, 1)$, $i = 1, \dots, n$. Here α and β are unknown parameters, and $x_1, \dots, x_n$ are known constants. Determine the uniformly most powerful unbiased test of $H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$.
- *5. Consider a regression model in which data $Y_1, \dots, Y_n$ are independent with $Y_i \sim N(\beta x_i + \gamma w_i, 1)$, $i = 1, \dots, n$. Here β and γ are unknown parameters, and $x_1, \dots, x_n$ and $w_1, \dots, w_n$ are known constants. Determine the uniformly most powerful unbiased test of $H_0 : \beta \leq \gamma$ versus $H_1 : \beta > \gamma$.
- *6. Let $X_1, \dots, X_m$ be a random sample from the Poisson distribution with mean λ_x , and let $Y_1, \dots, Y_n$ be an independent random sample from the Poisson distribution with mean λ_y .
 - a) Describe the uniformly most powerful unbiased test of $H_0 : \lambda_x \leq \lambda_y$ versus $H_1 : \lambda_x > \lambda_y$.
 - b) Suppose $\alpha = 5\%$ and the observed data are $X_1 = 3$, $X_2 = 5$, and $Y_1 = 1$. What is the chance the uniformly most powerful test will reject H_0 ?
 - c) Give an approximate version of the test, valid if m and n are large.
7. Consider a two-parameter exponential family with density

$$p_{\theta, \eta}(x, y) = \begin{cases} \frac{\theta^2 \eta^2 (x + y)}{\theta + \eta} e^{-\theta x - \eta y}, & x > 0 \text{ and } y > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Determine the level α uniformly most powerful unbiased test of $H_0 : \theta \leq \eta$ versus $H_1 : \theta > \eta$.

8. Suppose X and Y are absolutely continuous with joint density

$$p_{\theta}(x, y) = \begin{cases} (x + y)e^{-\theta_1 x^2 - \theta_2 y^2 - A(\theta)}, & x > 0, y > 0; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find $A(\theta)$.
 - b) Find the uniformly most powerful unbiased test of $H_0 : \theta_1 \leq \theta_2$ versus $H_1 : \theta_1 > \theta_2$.
9. Let X have a normal distribution with mean μ and variance σ^2 .
- a) For $x > 0$, find

$$p(x) = \lim_{\epsilon \downarrow 0} P(x < X < x + \epsilon \mid x^2 < X^2 < (x + \epsilon)^2).$$

In part (b) you can assume that the conditional distribution of X given X^2 is given by $P(X = x \mid X^2 = x^2) = p(x)$ and $P(X = -x \mid X^2 = x^2) = 1 - p(x)$, $x > 0$.

- b) Find the uniformly most powerful unbiased level α test φ of $H_0 : \mu \leq 0$ versus $H_1 : \mu > 0$ based on the single observation X .
10. Let $X_1, \dots, X_n$ be i.i.d. from $N(\mu, \sigma^2)$. Determine the uniformly most powerful unbiased test of $H_0 : \sigma^2 \leq 1$ versus $H_1 : \sigma^2 > 1$.

